

Multiple Choice (10 marks)

- Which of the following is an n-type semiconductor?
 - Silicon
 - Diamond
 - Silicon carbide
 - Arsenic-doped silicon
- Calculate the polarization of a proton in a magnetic field of 335 mT and 10.5 T at 298 K.
 - 6.345×10^{-4} at 0.335 T; 9.871×10^{-5} at 10.5 T
 - 0.793×10^{-4} at 0.335 T; 6.931×10^{-7} at 10.5 T
 - 1.148×10^{-6} at 0.335 T; 3.598×10^{-5} at 10.5 T
 - 4.126×10^{-3} at 0.335 T; 2.142×10^{-6} at 10.5 T
- Which of the following is lower for argon than for neon?
 - Melting point
 - Boiling point
 - Polarizability
 - First ionization energy
- Of the following atoms, which has the lowest electron affinity?
 - F
 - Si
 - O
 - Ca
- Of the following compounds, which has the lowest melting point?
 - HCl
 - AgCl
 - CaCl₂
 - CCl₄

True / False (10 marks)

Answer True or False

6. True or False: In the balanced equation, the ratio of MnO_4^- to I^- is 6:5.
7. True or False: Raman spectroscopy is a light-scattering technique used to study molecular vibrations.
8. True or False: CH_3Cl exhibits a ^1H resonance approximately 1.55 kHz downfield from TMS on a 12.0 T NMR spectrometer.
9. True or False: A magnetic field of 6.08 T is optimal for achieving the fastest spin-lattice relaxation in protons of a molecule with a 1 ns correlation time.
10. True or False: The ^{55}Mn nuclide has three allowed energy levels for its nuclear states.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that determine the ratio of line intensities in the EPR spectrum of the t-Bu radical $(\text{CH}_3)_3\text{C}^\bullet$, and explain the significance of this ratio in understanding radical behavior.
12. Discuss the factors that contribute to the NMR frequency of nuclides, and explain why ^{17}O exhibits a frequency of 115.5 MHz in a 20.0 T magnetic field.

End of Paper